Assignment – 2

Topic: Pattern Visualization & Filtering the Square

Instructions:

1. Take input `n` from the user wherever required.
2. Do a **dry run on paper first** before coding.
3. **Identify:**
    1. **What does the outer loop represent?**
    2. **What does the inner loop represent?**
4. **Do NOT copy from chatGPT / friends — focus on understanding.**

Q1) Hollow Square

Print a hollow square of size `n`.

Example

If `n = 5`

```
*   *   *   *   *
*               *
*               *
*               *
*   *   *   *   *
```

Think:

Which rows are fully filled?

Which columns are fully filled?

What should be printed in the middle?

Q2) Diagonal Square (X Pattern)

Print the following pattern:

If `n = 5`

```
*           *
    *   *
        *
    *   *
*           *
```

Think:

When are row and column equal?

When does row + column equal something constant?

Q3) Right Triangle (Derived from Square)

If `n = 5`

```
*
*   *
*   *   *
*   *   *   *
*   *   *   *   *
```

Think:

In square, inner loop runs till `n`

What condition makes it triangular?

When should printing stop in each row?

Q4) Inverted Right Triangle

If `n = 5`

```
*   *   *   *   *
*   *   *   *
*   *   *
*   *
*
```

Think:

How does number of columns change as rows increase?

What relationship exists between `i` and `n`?

Q5) Chessboard Pattern

If `n = 4`

```
*   #   *   #
#   *   #   *
*   #   *   #
#   *   #   *
```

`Submit:`

1. Code
2. Dry run for n = 3
3. Short explanation:
    - What does outer loop represent?
    - What does inner loop represent?

YENJOYY !!